ORF307
Sensitivity analysis

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Today's topics

- Primal and dual simplex
- Adding variables and constraints
- Global sensitivity
- Local sensitivity

- **Recap**

Optimal objective values

Primal

$$\min c^T x$$
$$\text{s.t. } Ax \le b$$

Dual

$$\max_{y} \ -b^T y$$
$$\text{s.t. } A^T y + c = 0$$
$$y \ge 0$$

$\boldsymbol{p}^*$ is the primal optimal value

$\boldsymbol{d}^*$ is the primal optimal value

Infeasible Primal → $p^* = +\infty$

Unbounded Primal → $p^* = -\infty$

Infeasible Dual → $d^* = -\infty$

Unbounded Dual → $d^* = +\infty$

Weak duality

Theorem: if $x \in R^n$ and $y \in R^m$ satisfy

- x is a feasible solution to the Primal
- y is a feasible solution to the Dual

$$\left.\begin{array}{l} x \text{ is a feasible solution to the Primal} \\ y \text{ is a feasible solution to the Dual} \end{array}\right\} \Rightarrow \quad -b^T y \le c^T x$$

Proof:

$$\left.\begin{array}{l} \text{Since } x \text{ is feasible for the Primal: } Ax \le b \Rightarrow y^T Ax \le y^T b. \\ \text{Since } y \text{ is feasible for the Dual: } Ay + c = 0 \text{ and } y \ge 0. \end{array}\right\} \Rightarrow y^T Ax \le y^T b \Rightarrow -c^T x \le b^T y \Rightarrow -b^T y \le c^T x$$

$\blacksquare$

Remarks:

- For any dual feasible solution y, the dual objective $-b^T y$ is a **lower bound** to the primal objective $c^T x$
- For any primal feasible solution x, the primal objective $c^T x$ is an **upper bound** to the dual objective $-b^T y$
- **Duality gap**: $g = c^T x + b^T y$

Strong duality

Theorem:

If an LP has an optimal solution x^* then its dual LP has also an optimal solution y^*, and the optimal objective values of the Primal and Dual LPs are equal, i.e.,

$$\boldsymbol{p}^* = c^T x^* = -b^T x = \boldsymbol{d}^*$$

In such a case the duality gap is zero.

Proof: see last lecture slides

Complementarity slackness

Primal:

$\min\ c^T x$
s.t. $Ax \leq b$

Dual:

$max\ -b^T y$
s.t. $A^T y + c = 0$
$y \geq 0$

Theorem:

Let x be primal feasible and y be dual feasible. x, y are optimal if an only if

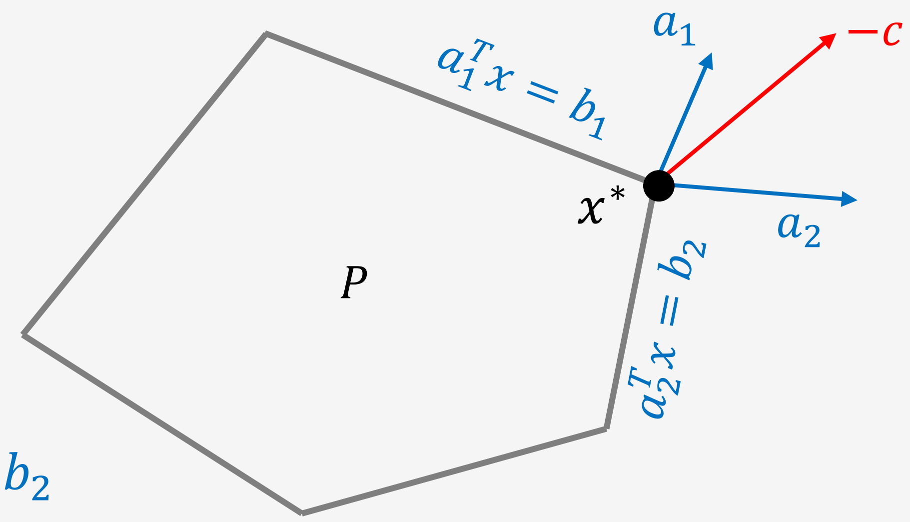
$$y_i\left(b - a_i^T x\right) = 0, \qquad i = 1,2, \dots, m$$

Remark: at an optimum (primal or dual), $b - Ax$ and y have **complementary sparsity pattern**:

$$y_i > 0 \ \Rightarrow\ a_i^T x - b = 0$$

$$a_i^T \mathrm{x} < \mathrm{b_i} \ \Rightarrow\ \mathrm{y_i} = 0$$

Geometric interpretation



Two active constraints at the **optimum**: $a_1^T x^* = b_1$ and $a_2^T x^* = b_2$

The **dual optimal solution** y satisfies:

$$A^T y + c = 0,\ y \geq 0, \qquad \textbf{and} \quad y_i = 0,\ \text{for}\ i \neq 1{,}2$$

Hence: $A^T y + c = 0 \Rightarrow a_1 y_1 + a_2 y_2 = -c$ **since only** $y_1, y_2 \geq 0$

$-c$ is in the cone of columns of A

$$A = \begin{bmatrix} a_1^T \\ \vdots \\ a_m^T \end{bmatrix}$$

Note: a_i^T is the i-th row of A and for A^T we have a_i as the i-th column

$$A^T = [a_1 \quad \cdots \quad a_m]$$

Lagrangian function and duality

Primal:

$$\min \ c^T x$$
$$\text{s.t.} \ \ Ax \le b$$

Dual:

$$max \ -b^T y$$
$$\text{s.t.} \ \ A^T y + c = 0$$
$$y \ge 0$$

Dual function:

$$g(y) = \min_x \left(c^T x + y^T (Ax - b) \right)$$

$$= -b^T y + \min_x (c + A^T y)^T x =$$

$$= \begin{cases} -b^T y, & \text{if } \ c + A^T y = 0 \\ -\infty & \text{, otherwise} \end{cases}$$

Lagrangian function

$$L(x, y) = c^T x + y^T (Ax - b)$$

NOTE: $\nabla_x L(x, y) = 0 \Leftrightarrow c + A^T y = 0$

KKT conditions

Optimality conditions of LP

Primal:

$\min\ c^T x$
s.t. $Ax \le b$

Dual:

$max\ -b^T y$
s.t. $A^T y + c = 0$
$y \ge 0$

Primal feasibility: $Ax \le b$

Dual feasibility: $A^T y + c = 0$, $y \ge 0$

Complementarity slackness: $y_i(Ax - b)_i = 0, i = 1, \ldots m$

KKT conditions

Optimality conditions of LP

Primal:

$$\min\ c^T x \quad \text{s.t. } Ax \leq b$$

Dual:

$$\begin{aligned} &max\ -b^T y \\ &\text{s.t. } A^T y + c = 0 \\ &\quad\ \ y \geq 0 \end{aligned}$$

We can solve our LP by solving the **<u>system of linear equations and inequalities</u>**

$$\nabla_x L(x, y) = A^T y + c = 0$$

$$y \geq 0$$

$$b - Ax \geq 0$$

$$y^T (b - Ax) = 0$$

- **Primal and dual simplex**

Primal and Dual basic feasible solutions

Optimality conditions of the standard form LP

Primal:

$\min\ c^T x$

s.t. $Ax = b$

$x \geq 0$

Dual:

$max\ -b^T y$

s.t. $A^T y + c \geq 0$

Primal feasibility: $Ax = b, x \geq 0$

Dual feasibility: $A^T y + c = 0$,

Duality gap is zero: $c^T x + b^T y = 0$

Primal and Dual basic feasible solutions

Optimality conditions of the standard form LP

Primal:

$\min\ c^T x$

s.t. $Ax = b$

$x \geq 0$

Dual:

$max\ -b^T y$

s.t. $A^T y + c \geq 0$

Primal feasibility: $Ax = b, x \geq 0 \ \Rightarrow\ x_B = A_B^{-1} b \geq 0$ (we set $x_N = 0$, N: set of non-basic variables)

Dual feasibility: $A^T y + c = 0 \Rightarrow A_B^T y = -\mathrm{c_B} \Rightarrow y = (A_B^T)^{-1} c_B$. If $\bar{c} = c + A^T y \geq 0 \ \Rightarrow\ y$ is dual feasible.

Question: Are $x_B = A_B^{-1} b$, $x_N = 0$ **and** $y = -A_B^{-T} c_B$ **optimal ??**

Primal and Dual basic feasible solutions

Optimality conditions of the standard form LP

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Dual feasibility: $A^T y + c = 0 \Rightarrow A_B^T y = -\mathrm{c_B} \Rightarrow y = (A_B^T)^{-1} c_B$. If $\bar{c} = c + A^T y \geq 0 \;\Rightarrow\; y$ is dual feasible.

Question: Are $x_B = A_B^{-1} b$, $x_N = 0$ **and** $y = -A_B^{-T} c_B$ **optimal ??**

They are optimal if the duality gap is zero.

Duality gap is zero: $c^T x + b^T y = c_B^T x_B + b^T(-A_B^T c_B) = c_B^T x_B - c_B^T A_B^{-1} b = c_B^T x_B - c_B^T x_B = 0$

Primal:

$$\min\ c^T x$$
$$\text{s.t. } Ax = b$$
$$x \geq 0$$

Dual:

$$max\ -b^T y$$
$$\text{s.t. } A^T y + c \geq 0$$

Primal Simplex
↓
Find initial BFS
↓
Iterate: maintain **Primal** feasibility
↓
At optimal vertex: Duality Gap=0
↓
Dual optimal solution found via complementarity

Dual Simplex
↓
Find initial BFS
↓
Iterate: maintain **Dual** feasibility
↓
At optimal vertex: Duality Gap=0
↓
Primal feasibility satisfied → both optimal

- If it is **easier** to find (compute) an BFS of the **Dual** problem, we should **start** by solving the dual. At the optimal solution of the Dual we can use **complementarity slackness** to compute the **optimal** solution of the **Primal problem.**
- Also, if we solve the **Primal** problem at its optimal solution we can compute the **dual variables** which give us the **shadow prices of the resources** we have in our Primal (original) problem.

In the Precepts we have this very useful table about the Primal Simplex and the Dual Simplex

	Primal Feasibility	Dual Feasibility	Duality Gap
Primal Simplex	✓ (yes)	× (violates until optimal)	✓ (yes)
Dual Simplex	× (violates until optimal)	✓ (yes)	✓ (yes)

Sensitivity Analysis

- **Adding new variables**

Adding new variables (columns)

Primal LP:

$\min\ c^T x$
s.t. $Ax = b$
$x \geq 0$

Optimal Solution: $\boldsymbol{x}^*$

Dual LP:

$max\ -b^T y$
s.t. $A^T y + c \geq 0$

Optimal Solution: $\boldsymbol{y}^*$

Primal LP with the new variable:

$\min\ c^T x + c_{n+1} x_{n+1}$
s.t. $Ax + A_{n+1} x_{n+1} = b$
$x \geq 0,\ x_{n+1} \geq 0$

Basic Feasible Solution: $(\boldsymbol{x}^*, \boldsymbol{0})$ (easily found)

Dual LP of the LP with the new variable :

$max\ -b^T y$
s.t. $A^T y + c \geq 0$
$A_{n+1}^T y + c_{n+1} \geq 0$

$\boxed{\boldsymbol{y}^* \text{ may be feasible or not} \rightarrow \text{last constraint is critical}}$

Are the solutions $(x^*, 0)$ and y^* optimal for the new LPs?

Adding new variables (columns)

Primal LP with the new variable:

$\min\ c^T x + c_{n+1} x_{n+1}$
s.t. $Ax + A_{n+1} x_{n+1} = b$
$x \geq 0,\ x_{n+1} \geq 0$

Basic Feasible Solution: $(\boldsymbol{x}^*, \boldsymbol{0})$

Check the reduced cost of the new variable x_{n+1}

$$\bar{c}_{n+1} = c_{n+1} - A^T (A_B^T)^{-1} c_B$$

IF $\bar{c}_{n+1} \geq 0$ **THEN** $(x^*, 0)$ is <u>still optimal</u> for the new LP
ELSE use $(x^*, 0)$ & <u>Primal Simplex</u> to compute new optimal

Dual LP of the LP with the new variable :

$max\ -b^T y$
s.t. $A^T y + c \geq 0$
$A_{n+1}^T y + c_{n+1} \geq 0$

$\boldsymbol{y}^*$ may be feasible or not → last constraint is critical

We know: $A^T y^* + c \geq 0$
IF $A_{n+1}^T y^* + c_{n+1} \geq 0$ **THEN** y^* is <u>still dual optimal</u> of new LP and strong duality $\Rightarrow$ $(x^*, 0)$ optimal of new LP
ELSE use $(x^*, 0)$ & Primal Simplex to compute new optimal

Are the solutions $(\boldsymbol{x}^*, \boldsymbol{0})$ and $\boldsymbol{y}^*$ optimal for the new LPs?

Example – *Adding new variable (column)*

$$\begin{aligned}
&\textbf{min} -60x_1 - 30x_2 - 20x_3 && (-)\textbf{profit}\\
&\text{s.t.}\quad 8x_1 + 6x_2 + x_3 \le 48 && \textbf{material}\\
&\qquad 4x_1 + 2x_2 + 1.5x_3 \le 20 && \textbf{production}\\
&\qquad 2x_1 + 1.5x_2 + 0.5x_3 \le 8 && \textbf{quality control}\\
&\qquad x_1, x_2, x_3 \ge 0
\end{aligned}$$

Primal LP:

$$\begin{aligned}
&\min\ c^T x\\
&\text{s.t.}\ \ Ax = b\\
&\qquad x \ge 0
\end{aligned}$$

$$c = (-60, -30, -20, 0, 0, 0)^T, \quad b = (48, 20, 8)^T$$

$$A = \begin{bmatrix} 8 & 6 & 1 & 1 & 0 & 0 \\ 4 & 2 & 1.5 & 0 & 1 & 0 \\ 2 & 1.5 & 0.5 & 0 & 0 & 1 \end{bmatrix}$$

We have added Slack variables

$x^* = (2,0,8,24,0,0)^T$, $y^* = (0,10,10)^T$, $c^T x^* = -280$, optimal basis $\mathrm{B} = \{1,3,4\}$

2nd and 3rd constraints are active

Example - Adding new variable (column)

Add a new product: $c = (-60, -30, -20,0,0,0, -15)^T$, $b = (48,20,8)^T$

$$\begin{aligned} \min\ & c^T x + c_{n+1}x_{n+1} \\ \text{s.t.}\ & Ax + A_{n+1}x_{n+1} = b \\ & x \geq 0, x_{n+1} \geq 0 \end{aligned}$$

$$A = \begin{bmatrix} 8 & 6 & 1 & 1 & 0 & 0 & 1 \\ 4 & 2 & 1.5 & 0 & 1 & 0 & 1 \\ 2 & 1.5 & 0.5 & 0 & 0 & 1 & 1 \end{bmatrix}$$

Previous solution:

$x^* = (2,0,8,24,0,0)^T$ $\quad y^* = (0,10,10)^T$ $\quad c^T x^* = -280$ $\quad$ basis $\ \mathrm{B} = \{1,3,4\}$

Is $(x^*, 0)$ is optimal?? Let's use strict duality to see if y^* is still optimal for the new dual

$$A_{n+1}^T y^* + c_{n+1} = [1,1,1]^T \begin{bmatrix} 0 \\ 10 \\ 10 \end{bmatrix} + (-15) = 5 \geq 0 \ \Rightarrow \ y^* \text{ is still optimal.}$$

Hence: $(x^*, 0)$ is optimal **Shall we add the new product?**

- **Adding new constraints**

Adding new constraint (row)

Primal LP

$\min\ c^T x$

s.t. $Ax = b$

$x \geq 0$

$\boldsymbol{x}^*$ is primal optimal

Primal LP with new constraint

$\min\ c^T x$

s.t. $Ax = b$

$\boldsymbol{a_{m+1}^T x = b_{m+1}}$

$x \geq 0$

$\boldsymbol{x}^*$ may be feasible or not → the new constraint is critical

Strong Duality: $\boldsymbol{c^T x^* = -b^T y^*}$

Dual LP:

$max\ -b^T y$

s.t. $A^T y + c \geq 0$

Optimal Solution: $\boldsymbol{y}^*$

Dual the LP with new constraint:

$\min\ -b^T y - \boldsymbol{b_{n+1} y_{n+1}}$

s.t. $A^T y + \boldsymbol{a_{m+1} y_{m+1}} + c \geq 0$

$(\boldsymbol{y}^*, \boldsymbol{0})$ is dual feasible

Are the solutions x^* and $(y^*, 0)$ optimal for the new LPs?

Adding new constraint (row)

Primal LP with new constraint

$\min\ c^T x$

s.t. $Ax = b$

$a_{m+1}^T x = b_{m+1}$

$x \geq 0$

$\boldsymbol{x}^*$ may be feasible or not → new constraint is critical

Dual the LP with new constraint:

$\max\ -b^T y + b_{n+1} y_{n+1}$

s.t. $A^T y + a_{m+1} y_{m+1} + c \geq 0$

$(\boldsymbol{y}^*, \mathbf{0})$ is dual feasible

IF $a_{m+1}^T x^* = b_{m+1}$ **THEN** x^* is **still** optimal for new LP

Also, $(\boldsymbol{y}^*, \mathbf{0})$ is dual feasible for new dual LP
From Strong Duality:

$$\boldsymbol{c}^T \boldsymbol{x}^* = -\boldsymbol{b}^T \boldsymbol{y}^* \Rightarrow \boldsymbol{c}^T \boldsymbol{x}^* = -\boldsymbol{b}^T \boldsymbol{y}^* - \boldsymbol{b}_{m+1} \mathbf{0}$$

Hence: $(y^*, 0)$ optimal for new dual LP.

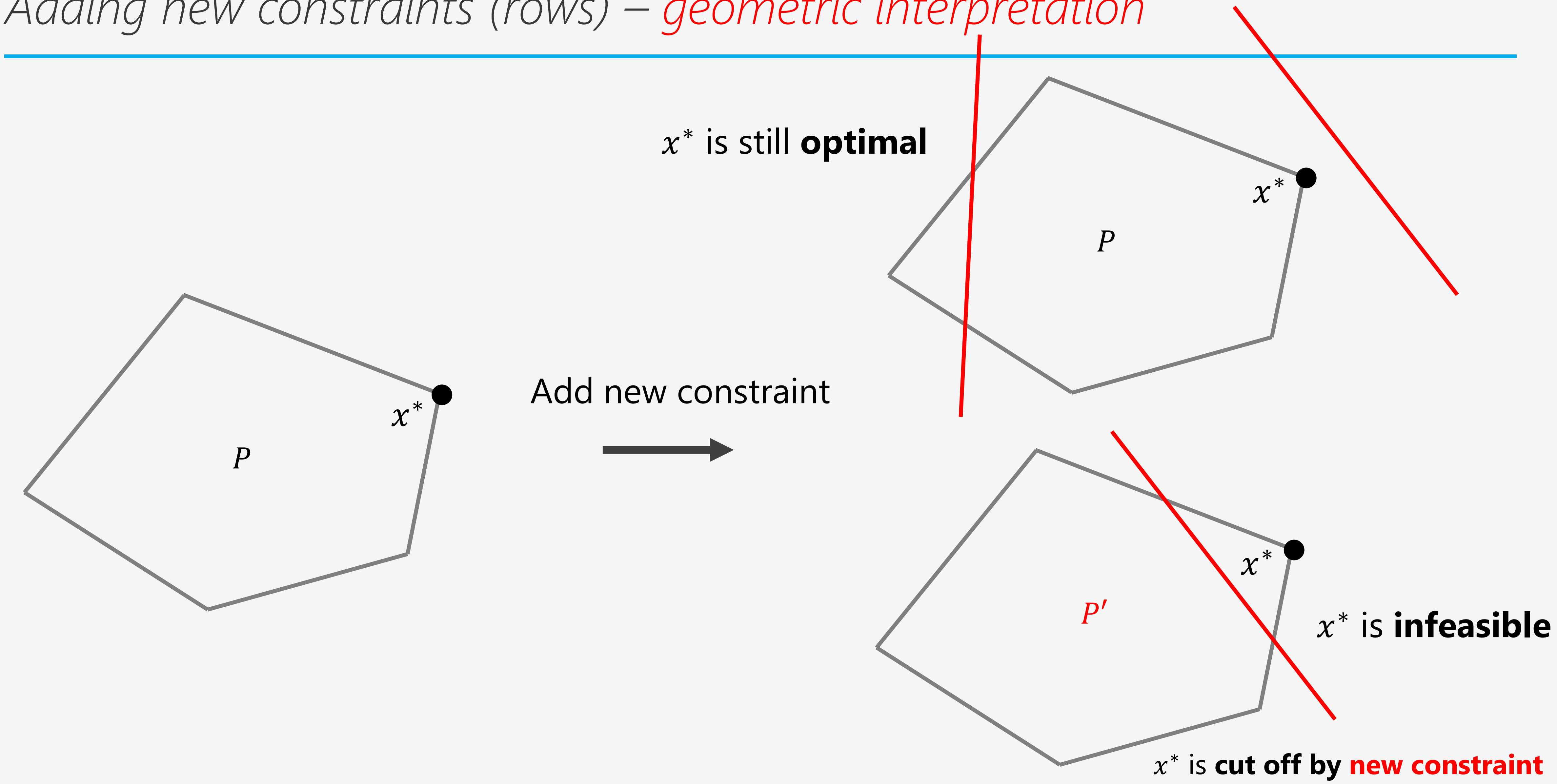
ELSE $\boldsymbol{x}^*$ is infeasible for new LP $(a_{m+1}^T x^* \neq b_{m+1})$

We can use **Dual Simplex** starting from $(\boldsymbol{y}^*, \mathbf{0})$ to converge to an optimal dual solution.

Then we **recover** the optimal of the new LP, say x^{**}, using **complementarity**.

Are the solutions $\boldsymbol{x}^*$ and $(\boldsymbol{y}^*, \mathbf{0})$ optimal for the new LPs?

Adding new constraints (rows) – geometric interpretation



- **Global sensitivity analysis**

Changes in problem data

Goal: derive information from the optimal primal-dual solutions x^*, y^* about

their sensitivity when the **data of the LP changes**

Modified right-hand side (rhs) of the LP

$$\begin{aligned} &\min\ c^T x \\ &\text{s.t. } Ax = b + u \\ &\qquad x \geq 0 \end{aligned}$$

The optimal cost becomes a function of $u \rightarrow p^*(u)$

$\boldsymbol{u}$ is called the **perturbation vector** of the rhs of the constraints

Changes in problem data

Dual of modified LP

$$\max -(b+u)^T y$$

$$\text{s.t.} \quad A^T y + c \geq 0$$

Global sensitivity analysis

Given y^* that is a dual optimal solution for $u = 0$, then

$$p^*(\mathrm{u}) \underset{\text{weak duality}}{\geq} -(b+u)^T y^* = -\boldsymbol{b}^{\boldsymbol{T}} \boldsymbol{y}^* - u^T y^* \underset{\text{strong duality}}{=} \boldsymbol{c}^{\boldsymbol{T}} \boldsymbol{x}^* - u^T y^* = p^*(0) - u^T y^*$$

Hence for every $\boldsymbol{u}$ we have: $\quad \boldsymbol{p}^*(\boldsymbol{u}) \geq \boldsymbol{p}^*(\boldsymbol{0}) - \boldsymbol{u}^{\boldsymbol{T}} \boldsymbol{y}^*$

Provides a global lower bound on the cost $p^*(u)$ relative to the original cost $p^*(0)$ and the perturbation u

Global sensitivity analysis

Example:

Take $u = td$ for some fixed $d \in R^m$

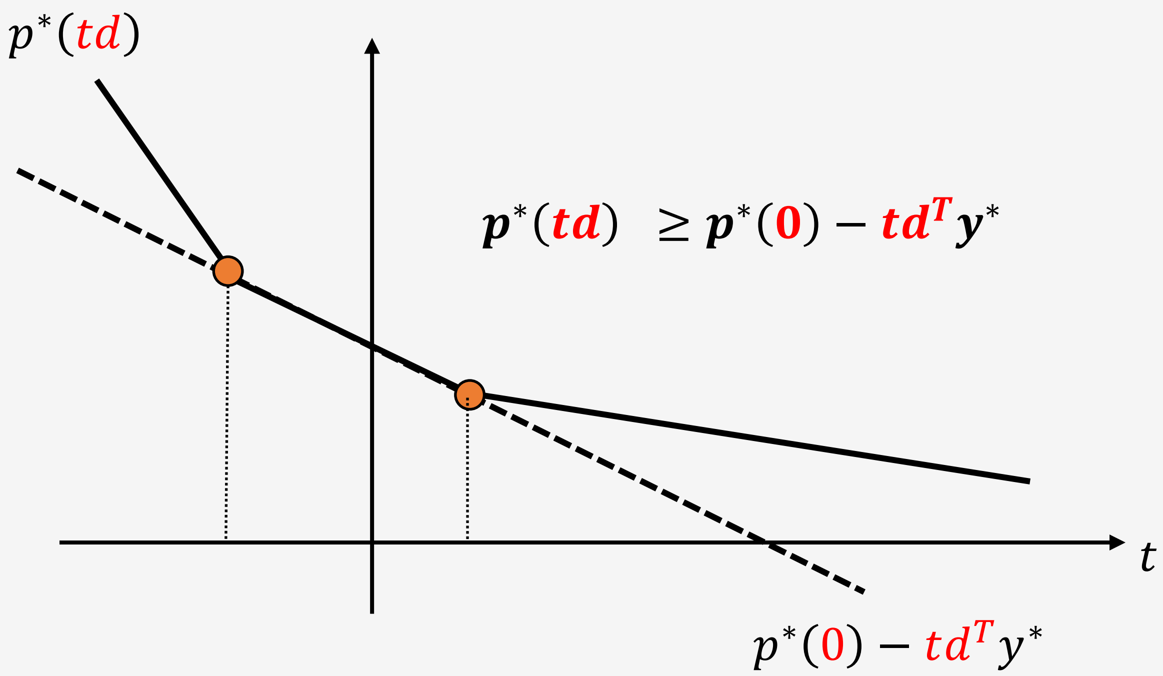
$\boxed{\text{*** We restrict } u \text{ to a specific direction } d}$

$\min c^T x$
s.t. $Ax = b + td$
$x \geq 0$

$p^*(td)$ is the optimal value as a **function** of t

Sensitivity information (assuming $d^T y^* \geq 0$)

- If $t < 0$ then the optimal value increases
- If $t > 0$ then the optimal value decreases



$p^*(td)$ is a <u>piecewise</u> linear function of t

Optimal value function is piecewise linear - Intuition

Let's see why $p^*(td)$ is a piecewise linear function of t

The optimal solution x^* corresponds to a **specific optimal basis** B and a **set of active constraints**

$$x_B = A_B^{-1}(b + td)$$

The **optimal objective value** is **linear in t** for that optimal basis B

$$\boldsymbol{p^*(td)} = c_B^T x_B = c_B^T A_B^{-1}(b + td) = c_B^T A_B^{-1} b + t c_B^T A_B^{-1} d = c_B^T A_B^{-1} b + t d^T y^* = \boldsymbol{p^*(0) + td^T y^*}$$

where $\boldsymbol{y^* = A_B^{-T} c_b}$ is the optimal **dual solution for the basis B**

$p^*(td)$ remains a single straight line until **t changes enough to make the basis B infeasible**
The **kinks** are caused by the change in the optimal basis from B to some other optimal basis $\bar{B}$

The **new linear segment** is now defined by the **new basis** $\bar{B}$

$$p^*(td) = c_{\bar{B}}^T x_{\bar{B}} = c_{\bar{B}}^T A_{\bar{B}}^{-1}(b + td) = c_{\bar{B}}^T A_{\bar{B}}^{-1} b + t c_{\bar{B}}^T A_{\bar{B}}^{-1} d = c_{\bar{B}}^T A_{\bar{B}}^{-1} b + t d^T \bar{y}^* = p^*(0) + t d^T \bar{y}^*$$

A series of line segments formed by varying t and the optimal basis, yields the **piecewise** linear function $p^*(td)$

Optimal value function is piecewise linear - Properties

$$p^*(u) = \min\{c^T x \mid Ax = b + u, \qquad x \geq 0\}$$

Assumption: $p^*(0)$ is finite (the original LP is feasible and bounded)

Properties:

- $p^*(u) > -\infty$ everywhere (from the global lower bound)
  - No unbounded LPs
- $p^*(u)$ is piecewise-linear on its domain

Optimal value function is piecewise linear

Proof:

We consider the optimal objective value: $p^*(u) = \min\{c^T x \mid Ax = b + u,\ x \geq 0\}$

Dual feasible set

$$D = \{y \mid A^T y + c \geq 0\}$$

Assumption: $p^*(0)$ is finite $\Rightarrow$ D has a **finite set of vertices** (r) $\Rightarrow$

$$\Rightarrow \ p^*(u) = \max_{y \in D} -(b+u)^T y = \max_{k=1,\ldots,r} \underbrace{\left(-y_k^T u - b^T y_k\right)}_{\substack{\text{Linear segment} \\ \text{Straight line}}}$$

Piecewise linear function

where $y_1, \ldots, y_r$ are the extreme points of D

- **Local sensitivity analysis**

Local sensitivity analysis

u is now in a neighborhood of the origin

Original LP:

$$\begin{aligned} \min\ & c^T x \\ \text{s.t.}\ & Ax = b \\ & x \geq 0 \end{aligned}$$

Optimal solution:

- **Primal**: $x_B^* = A_B^{-1} b$ and $x_i^* = 0,\ i \notin B$
- **Dual**: $y^* = -A_B^{-T} c_B$

Modified LP

$$\begin{aligned} \min\ & c^T x \\ \text{s.t.}\ & Ax = b + u \\ & x \geq 0 \end{aligned}$$

Modified Dual

$$\begin{aligned} \max\ & -(b+u)^T y \\ \text{s.t.}\ & A^T y + c \geq 0 \end{aligned}$$

Modified optimal solution:

$$x_B^*(u) = A_B^{-1}(b+u) = x_B^* + A_B^{-1} u$$

$$y^*(u) = y^*$$

The optimal basis does not change

Derivative of the optimal value function

Modified optimal solution:

$$x_B^*(u) = A_B^{-1}(b+u) = x_B^* + A_B^{-1}u$$

$$y^*(u) = y^*$$

Optimal value function:

$$p^*(u) = c^T x^*(u) = c^T x + c_B^T A_B^{-1} u$$

$$= p^*(0) - {y^*}^T u \quad \rightarrow \quad (\textbf{affine} \text{ for small } u)$$

Local derivative:

$$\nabla p^*(u) = -y^* \quad \rightarrow \quad (y^* \text{ are called } \textbf{shadow prices})$$

Shadow price (economic interpretation): The change in the objective function's optimal value (e.g., total profit or cost) for a one-unit increase in the constraint's availability

Example

$$\begin{aligned} &\min -60x_1 - 30x_2 - 20x_3 && \text{(−)profit} \\ &\text{s.t.}\quad 8x_1 + 6x_2 + x_3 \leq 48 && \text{material} \\ &\qquad 4x_1 + 2x_2 + 1.5x_3 \leq 20 && \text{production} \\ &\qquad 2x_1 + 1.5x_2 + 0.5x_3 \leq 8 && \text{quality control} \\ &\qquad x_1, x_2, x_3 \geq 0 \end{aligned}$$

$x^* = (2,0,8,24,0,0)^T$ $\quad y^* = (0,10,10)^T$ $\quad c^T x^* = -280$ $\quad$ basis $\mathrm{B} = \{1,3,4\}$

What does $y_3^* = 10$ really mean?

If we increase the quality control budget by 1 (i.e., we select $u = (0,0,1)^T$), the optimal objective value becomes:

$$p^*(u) = p^*(0) - y^{*T} u = -280 - 10 = -290$$

Profit increases by a known quantity
Managers (and CFOs) like this information!

References

- Bertsimas/Tsitsiklis: Introduction to Linear Optimization
  - Ch. 5: Sensitivity analysis
- Vanderbei: Linear Programming: Foundations and Extensions
  - Ch. 7: Sensitivity and parametric analysis

Next lecture

- Network optimization